10 ft Hex Cedar Gazebo — Master Plan

Location: Saanich, BC
Style: Traditional open-rafter cedar gazebo — mortise & tenon, no metal hangers
Inspiration: Natural cedar hexagonal gazebo; 6×6 posts on concrete pads, exposed rafter system with king post hub, pronounced curved knee braces
Revision: 0 | April 2025

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1. Project Overview

A full-size 10-foot hexagonal cedar gazebo with a 4:12 pitched hip roof converging at a central king post hub.
All joinery is traditional mortise and tenon secured with 3/4" white oak drawbore pegs.
No metal post hangers or joist connectors — in the tradition of timber-frame construction.

Key Specifications

Parameter	Value
Footprint	10'-0" point-to-point (5'-0" post radius)
Post spacing	5'-0" on-centre (60° intervals)
Total height at hub apex	10'-7" (10.601 ft)
Clearance at beam soffit	8'-4" (8.33 ft post height)
Beam depth	7-1/4" (4×8 nominal)
Roof pitch	4:12 (18.43°)
Hip rafter miter	**28.71°**
Hip rafter bevel	**9.10°**
Rafter overhang	12" past beam centreline
Posts	6×6 Western Red Cedar
All members	Western Red Cedar
Finish	Penofin Transparent Cedar, 2–3 coats
Foundation	12" dia. concrete caissons, 24" deep, J-bolt anchors
Location seismic / wind	Saanich BC — coastal wind ~25 psf; check E66 base spec with engineer

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2. Architectural Drawings

All drawings produced from locked geometry (`context/staging/structural-model.json`).
Open SVG files in any modern browser for full zoom and inspection.

Drawing	File	Status

Plan view (top down)	drawing-plan-view.svg	✅ VALIDATED
Front elevation	drawing-elevation-view.svg	✅ VALIDATED
Isometric projection	drawing-isometric-view.svg	✅ VALIDATED
Perspective view	drawing-perspective-view.svg	✅ VALIDATED

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3. Shop Blueprints

Dimensioned construction drawings with title blocks for shop use.

Sheet	File	Contents
SB-01 Plan	blueprint-plan.svg	Plan with post layout, dimensions, post-base callout
SB-02 Elevation	blueprint-elevation.svg	Elevation with compound-cut detail balloon
SB-03 Isometric	blueprint-isometric.svg	Assembly isometric with sequence numbers 1–7
SB-04 Details	blueprint-component-isolation.svg	4 joinery details: seat cut, tenon, post-beam, caisson

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4. Structural Members & Cut List

Full machine-readable list: `../output/shop-blueprint/SB01-cut-list.json`

ID	Member	Profile	Cut Length	Qty	Notes
M01	Main posts	6×6 WRC	8'-4"	6	Square ends; seal end grain
M02	Ring beam	4×8 WRC	5'-3"	6	Opposing 28.71° miters
M03	Hip rafter	4×6 WRC	5'-4" + 12" tail	6	Compound: 28.71° / 9.10°; bird's mouth at beam
M04	King post hub	6×6 WRC	2'-0"	1	6× mortises at 60° intervals
M05	Knee brace	4×4 WRC	3'-0"	12	45° compound bevel both ends
M06	Purlin ring	4×4 WRC	3'-3"	6	Half-lap to rafters mid-span

Total lumber: ~490 BF Western Red Cedar (including 10% waste)

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5. Joinery Details

All joints: traditional mortise and tenon, 3/4" white oak drawbore pegs. No metal hangers or joist connectors.

Joint	Detail Sheet	Mortise Size	Peg Count
Post top → beam	SB-04 Detail 3	2.125"×4.125"×3" blind	2 pegs
Rafter → hub	SB-04 Detail 2	1.625"×3.125"×3" blind	2 pegs
Knee brace → post and beam	SB-04 Detail 3	1.5"×3"×2.5" blind	2 pegs per end
Purlin half-lap → rafter	SB-04 Detail 1 note	Half-lap depth 1.5"	1 peg per joint

See SB-04 component isolation sheet for full dimensioned details.

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6. Compound Cut Reference

> ***Test on scrap before every production run.*** *These angles were computed by* `geometry_engine.py`.

Cut	Miter (fence)	Bevel (blade)	End
Hip rafter seat cut	28.71°	9.10°	Top/hub end
Hip rafter tail	0°	9.10°	Tail/overhang end
Ring beam ends	28.71°	0°	Both (opposing)
Knee brace ends	45°	0°	Both ends

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7. Foundation

Element	Spec
Caisson diameter	12"
Caisson depth	24" below grade
Frost depth (Saanich BC)	18" nominal
Anchor	5/8"×10" J-bolt galvanised, set in wet concrete
Post base	Simpson E66 standoff (no post-in-ground)
Concrete	Quikrete 80 lb High-Strength, 2 bags per caisson

Permit note: verify setback from property lines with City of Saanich (250-475-5671) before digging. Seismic zone 4 — confirm E66 hold-down spec with an engineer.

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8. Bill of Materials Summary

Document	File

Detailed cut list (JSON)	../output/shop-blueprint/SB01-cut-list.json
Lumber purchase list	lumber-purchase-list.md
Budget estimate (CAD)	budget-estimate.md

Budget range: $3,090 – $4,850 CAD (DIY labour; Saanich BC April 2025 pricing)
Mid-case estimate: $3,855 CAD

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9. Assembly Sequence

Full step-by-step guide: `assembly-guide.md`

Phase	Task	Crew
1	Site layout + caissons	1–2
2	Post preparation + shop cuts	1
3	Erect posts + beam ring	2 min.
4	Knee braces + rafters + hub + purlins	2–3
5	Final finish + inspection	1

Estimated build time: **4–6 days** for competent DIY crew of 2 (excludes curing time).

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10. Geometry Source

All angles, heights, and lengths derive from `plugins/garden-structure-designer/scripts/geometry_engine.py`.
Do not re-derive by hand — the engine handles the compound-angle trig correctly.
Locked model: `context/staging/structural-model.json` (`_locked: true`)

```
{
  "pitch": "4:12",
  "miter_deg": 28.71,
  "bevel_deg": 9.10,
  "post_cut_ft": 8.33,
  "total_height_ft": 10.601,
  "rafter_overhang_ft": 1.0
}
```